

# Short-Text Classification Based on ICA and LSA

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**Abstract.** Many applications, such as word-sense disambiguation and information retrieval, can benefit from text classification. Text classifiers based on Independent Component Analysis (ICA) try to make the most of the independent components of text documents and give in many cases good classification effects. Short-text documents, however, usually have little overlap in their feature terms and, in this case, ICA can not work well. Our aim is to solve the short-text problem in text classification by using Latent Semantic Analysis (LSA) as a data preprocessing method, then employing ICA for the preprocessed data. The experiment shows that using ICA and LSA together rather than only using ICA in Chinese short-text classification can provide better classification effects.

## 1 Introduction

Text classification (TC) according to documents' topics is an application of automatic analysis of documents in modern information processing systems. ICA applied on word context data can give distinct features which reflect syntactic and semantic categories [1]. The application of ICA in TC is based on the assumption that a document corpus is generated by a combination of several thematic topics. Thus, the independent components obtained by the ICA algorithms define statistically independent clusters of documents, allowing their thematic classification [2]. Text classifiers based on ICA, in many cases, give good classification effects. In short-text case, however, ICA can not work well because short-text documents usually have little overlap in their feature terms.

Our aim is to solve the short-text problem we called STP in text classification by using LSA as a data preprocessing method. ICA is then employed for the preprocessed data by utilizing its great classifying capability of identifying a generalizable low-dimensional basis set in the face of high-dimensional noisy data [3].

## 2 LSA and ICA Related to TC

TC methods are almost depended on document vector space model (VSM) [4] representation that our analyses and experiments are also based on.

Latent semantic analysis (LSA) [5] is a well-known simple approach to automatic generation of the concepts and analysis of terms co-occurrence that are useful in text classification and information retrieval purposes. LSA is based on singular value

decomposition (SVD) [6] by which a smaller document matrix  $A$  that retains only the most important information from the original document matrix can be constructed.

$$A = TSD^T. \quad (1)$$

In (1),  $T$  and  $D$  contain orthogonal vectors and  $S$  is diagonal matrix of singular values. The eigenvectors with the largest singular value capture the axes of largest variation in data. So each document can be projected into a lower dimensional space, which is called dimensionality reduction often used to reduce data noise, avoid overlearning. After getting the reduced space, same co-occurrence terms are projected on the same dimensionality and documents are represented as linear combinations of orthogonal features. [4], [6] have introduced the applications of LSA in TC. [5] describes LSA in detail.

ICA is a method for presenting a set of multivariate observations as a linear combination of unknown latent variables that are statistically independent, giving no information on the values of the other components [7]. Lately it has been found out that it is a suitable method for analyzing text documents [3], [8].

We suppose that there are some main topics acting as a set of independent hidden random variables in each document. ICA as a technique that exploits the more effective higher order statistical features in documents can be employed to determine these independent random variables and extract the independent topic structure of documents. The classic ICA model can be expressed as following:

$$x = As. \quad (2)$$

In (2),  $x$  is the documents topic-mixed information, a  $T \times D$  term by document matrix.  $A$  is a  $T \times N$  mixing matrix and  $S$  is a  $N \times D$  documents topic information matrix,  $N$  is the number of independent components representing main topics of a set of documents. If mixing matrix  $A$  is known, topics in a document corpus can be estimated as  $y = A^{-1}x = Wx$ , where separating matrix  $W$  can be determined by employing FastICA [7] algorithm. Consequently, new unseen documents can be classified under  $N$  topic categories by comparing their independent components with the ones of training documents, after projecting them to the ICA space by means of matrix  $W$  [2].

### 3 Short-Text Classification Problem

Each document has its special topic used for TC. Topics are considered as a set of independent hidden random variables. In VSM, suitable feature terms consisting of document vectors often implicate underlying topic of documents. Likewise, a topic can also determine which terms to appear in documents. If two documents, usually long-text documents, have some common feature terms, the same topic the two documents probably belong to. If regarding a set of documents as a topic-mixed corpus, TC based on ICA is then an application that ICA algorithm can maximize the statistical independence between documents from the observed topic-mixed document corpus and try to determine the documents' real topic categories.

But short-text documents usually have little overlap in their feature terms, especially after filtering stop-words or function-words. Fig.1 illustrates this case. This means that vectors of such documents are already independent of each other so that